

STACK (15 Questions)

Q1. (GATE CSE 2021 – Previous Year Style)

Consider the following C code:

```
#define MAX 5
int stack[MAX];
int top = -1;

void push(int x){
    if(top < MAX-1)
        stack[++top] = x;
}
```

After executing:

```
push(10);
push(20);
push(30);
```

What is the value of top?

- A) 0
- B) 1
- C) 2
- D) 3

Q2. (ISRO Scientist/Engineer 2020 – Previous Year Style)

Which data structure is used for recursive function calls?

- A) Queue
- B) Stack
- C) Linked List
- D) Heap

Q3. (BARC OCES 2022 – Previous Year Style)

The postfix expression

$AB+C^*$

is equivalent to

- A) $A+(B^*C)$
- B) $(A+B)^*C$
- C) $A^*(B+C)$
- D) $(A^*B)+C$

Q4. (DRDO CEPTAM 2021 – Previous Year Style)

Which operation has **$O(1)$** time complexity in an array implementation of stack?

- A) Search
- B) Push
- C) Traverse
- D) Sort

Q5. (HPCL Engineer 2019 – Previous Year Style)

Overflow in a stack occurs when

- A) Stack becomes empty
- B) Stack becomes full
- C) Top becomes zero
- D) Top becomes negative

Q6. (IOCL Engineer 2022 – Previous Year Style)

Which expression converts an infix expression into postfix?

- A) Queue
- B) Stack
- C) Heap
- D) Tree

Q7. (BEL Engineer 2021 – Previous Year Style)

For a stack implemented using linked list, insertion takes

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n \log n)$

Q8. (GATE CSE 2018 – Previous Year Style)

Which traversal of a binary tree can be implemented using a stack?

- A) Level Order
- B) Preorder
- C) Breadth First
- D) None

Q9. (ONGC Graduate Trainee 2023 – Previous Year Style)

If stack contains

10

20

30

(top = 30)

After one pop operation, top becomes

- A) 10
- B) 20
- C) 30
- D) NULL

Q10. (ECIL Technical Officer 2020 – Previous Year Style)

Which principle is followed by Stack?

- A) FIFO
- B) LIFO
- C) Random
- D) Circular

Q11. (GATE CSE 2019 – Previous Year Style)

Minimum number of stacks required to evaluate postfix expression?

- A) 2
- B) 1
- C) 3
- D) Depends on operands

Q12. (BARC OCES 2020 – Previous Year Style)

Balanced parentheses checking uses

- A) Queue
- B) Stack
- C) Graph
- D) Heap

Q13. (ISRO Scientist 2022 – Previous Year Style)

Pop operation deletes

- A) First element
- B) Middle element
- C) Last inserted element
- D) Random element

Q14. (DRDO CEPTAM 2019 – Previous Year Style)

Which pointer stores the top element in stack?

- A) head
- B) rear
- C) top
- D) front

Q15. (GATE CSE 2023 – Previous Year Style)

Maximum number of elements in a stack of size 10 is

- A) 9
- B) 10
- C) 11
- D) Depends on compiler

Q.No.	Correct Answer	Explanation
1	C	After three pushes, <code>top</code> becomes 2 (starts from -1).
2	B	Function recursion uses the Stack .
3	B	$AB+C* = (A+B) \times C$.
4	B	Push operation takes O(1) time.
5	B	Overflow occurs when the stack is full .
6	B	Infix-to-postfix conversion uses a Stack .
7	C	Insertion in a linked-list stack is O(1) .
8	B	Preorder traversal can be implemented using a stack.
9	B	After popping 30, the new top is 20 .
10	B	Stack follows LIFO .
11	B	One stack is sufficient for postfix evaluation.
12	B	Parentheses matching uses a Stack .
13	C	Pop removes the last inserted element.
14	C	The top pointer stores the top element's position.
15	B	A stack of size 10 stores 10 elements .

QUEUE (15 Questions)

Q1. (GATE CSE 2022 – Previous Year Style)

Queue follows

- A) LIFO
- B) FIFO
- C) Random
- D) Priority

Q2. (BARC OCES 2021 – Previous Year Style)

Insertion in queue is performed at

- A) Front
- B) Rear
- C) Middle
- D) Anywhere

Q3. (ISRO Scientist 2020 – Previous Year Style)

Deletion from queue occurs at

- A) Rear
- B) Front
- C) Middle
- D) Random

Q4. (BEL Engineer 2019 – Previous Year Style)

Circular queue is mainly used to

- A) Save memory
- B) Eliminate false overflow
- C) Increase complexity
- D) Reduce speed

Q5. (DRDO CEPTAM 2022 – Previous Year Style)

Time complexity of enqueue operation

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

Q6. (HPCL Engineer 2021 – Previous Year Style)

Which application uses Queue?

- A) Function calls
- B) CPU Scheduling
- C) Expression evaluation
- D) Recursion

Q7. (GATE CSE 2020 – Previous Year Style)

Which pointer changes during enqueue?

- A) Front
- B) Rear
- C) Top
- D) Head

Q8. (IOCL Engineer 2023 – Previous Year Style)

Queue overflow occurs when

- A) Queue empty
- B) Queue full
- C) $\text{Front} = \text{Rear}$
- D) $\text{Front} = -1$

Q9. (ECIL Technical Officer 2022 – Previous Year Style)

Priority Queue removes

- A) First inserted
- B) Highest priority
- C) Last inserted
- D) Random

Q10. (BARC OCES 2019 – Previous Year Style)

Deque stands for

- A) Double Ended Queue
- B) Dynamic Queue
- C) Double Queue
- D) Directed Queue

Q11. (GATE CSE 2021 – Previous Year Style)

Which traversal uses Queue?

- A) DFS
- B) BFS
- C) Inorder
- D) Postorder

Q12. (ONGC GT 2020 – Previous Year Style)

Front pointer indicates

- A) Last element
- B) First element
- C) Middle
- D) NULL

Q13. (ISRO Scientist 2023 – Previous Year Style)

Worst case enqueue complexity in linked queue

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n^2)$

Q14. (BEL Engineer 2020 – Previous Year Style)

Which queue removes element from both ends?

- A) Priority Queue
- B) Circular Queue
- C) Dequeue
- D) Linear Queue

Q15. (GATE CSE 2019 – Previous Year Style)

Circular Queue uses

- A) Modulo Arithmetic
- B) Recursion
- C) Dynamic Programming
- D) Hashing

Q.No.	Correct Answer	Explanation
1	B	Queue follows FIFO .
2	B	Insertion is performed at the rear .
3	B	Deletion is performed from the front .
4	B	Circular queues eliminate false overflow .
5	A	Enqueue operation is O(1) .
6	B	CPU Scheduling commonly uses a Queue .
7	B	During enqueue, the rear pointer changes.
8	B	Overflow occurs when the queue is full .
9	B	A priority queue removes the highest-priority element.
10	A	Deque means Double Ended Queue .
11	B	BFS uses a Queue .
12	B	The front pointer indicates the first element .
13	A	Enqueue in a linked queue takes O(1) .
14	C	A Deque allows insertion/deletion at both ends.
15	A	Circular queues use modulo arithmetic .

LINKED LIST (15 Questions)

Q1. (GATE CSE 2023 – Previous Year Style)

Which data structure uses dynamic memory allocation?

- A) Array
- B) Linked List
- C) Stack
- D) Queue

Q2. (BARC OCES 2021 – Previous Year Style)

A node in singly linked list contains

- A) Data only
- B) Pointer only
- C) Data and Pointer
- D) Index only

Q3. (ISRO Scientist 2022 – Previous Year Style)

Insertion at beginning of linked list takes

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n^2)$

Q4. (HPCL Engineer 2020 – Previous Year Style)

Deleting first node requires

- A) Updating head pointer
- B) Updating tail pointer
- C) Sorting list
- D) None

Q5. (GATE CSE 2018 – Previous Year Style)

Linked list overcomes which drawback of arrays?

- A) Random Access
- B) Fixed Size
- C) Searching
- D) Sorting

Q6. (BEL Engineer 2022 – Previous Year Style)

Traversal of linked list requires

- A) Index
- B) Pointer
- C) Queue
- D) Stack

Q7. (DRDO CEPTAM 2023 – Previous Year Style)

Last node pointer contains

- A) Zero
- B) NULL
- C) Address of Head
- D) Data

Q8. (IOCL Engineer 2022 – Previous Year Style)

Circular linked list last node points to

- A) NULL
- B) First Node
- C) Middle Node
- D) Previous Node

Q9. (GATE CSE 2020 – Previous Year Style)

Searching in singly linked list requires

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n^2)$

Q10. (BARC OCES 2019 – Previous Year Style)

Doubly linked list contains

- A) One pointer
- B) Two pointers
- C) Three pointers
- D) No pointer

Q11. (ONGC GT 2023 – Previous Year Style)

Memory utilization in linked list is

- A) Static
- B) Dynamic
- C) Fixed
- D) Constant

Q12. (ISRO Scientist 2021 – Previous Year Style)

Which operation is difficult in singly linked list?

- A) Forward Traversal
- B) Backward Traversal
- C) Insertion at Beginning
- D) Deletion at Beginning

Q13. (BEL Engineer 2023 – Previous Year Style)

Head pointer stores

- A) Last node
- B) First node address
- C) Middle node
- D) Data

Q14. (GATE CSE 2022 – Previous Year Style)

Loop detection in linked list commonly uses

- A) Binary Search
- B) Floyd's Cycle Detection Algorithm
- C) Merge Sort
- D) BFS

Q15. (DRDO CEPTAM 2020 – Previous Year Style)

Time complexity of inserting at the end of a singly linked list (without a tail pointer) is

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n^2)$

Q.No.	Correct Answer	Explanation
1	B	Linked lists use dynamic memory allocation .
2	C	A node stores data and a pointer .
3	A	Insertion at the beginning takes $O(1)$.
4	A	Delete the first node by updating the head pointer .
5	B	Linked lists overcome the fixed-size limitation of arrays.
6	B	Traversal is performed using pointers .
7	B	The last node's next pointer is NULL .
8	B	In a circular linked list, the last node points to the first node .
9	C	Linear search in a linked list takes $O(n)$.
10	B	A doubly linked list contains two pointers (<code>prev</code> and <code>next</code>).
11	B	Linked lists use dynamic memory.
12	B	Backward traversal is difficult in a singly linked list.
13	B	The head pointer stores the address of the first node .
14	B	Floyd's Cycle Detection Algorithm is used to detect loops.
15	C	Without a tail pointer, insertion at the end requires traversal, so it is $O(n)$.